Homework #10

This is the homework for week 11. The due date is 11:59 p.m., December 8. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (35 points) Suppose data $X_1, X_2, \cdots, X_n$ are i.i.d. normal $N(\mu, \sigma^2)$ random variables with μ unknown and σ^2 known. Suppose we want to test the hypothesis

$$H_0 : \mu \geq \mu_0 \text{ versus } H_1 : \mu < \mu_0,$$

with μ_0 a known number.

i. Write down the likelihood function $L(\mu)$ and log-likelihood function $\ell(\mu)$.

ii. Calculate the maximizer $\arg\max_{\mu \geq \mu_0} \ell(\mu)$.

iii. We consider the likelihood ratio test: reject H_0 if

$$\frac{\sup_{\mu \geq \mu_0} L(\mu)}{\sup_{\mu \in \mathbf{R}} L(\mu)} \leq c$$

for some constant c. Prove that this is equivalent to: reject H_0 if

$$\frac{1}{n} \sum_{i=1}^{n} X_i < C$$

for some constant C (which may not equal c).

iv. Suppose we want to control type-I error to be α, derive the constant C.

(**Question2):** (35 points) Suppose we have a set of data $Y_1, Y_2, \cdots, Y_n$, which are independent and satisfying

$$Y_i \sim N(x_i \beta_1 + \beta_0, \sigma^2), \text{ where } x_i \text{ is known.}$$

i. Write down the $\log-$likelihood function $\ell(\beta_0, \beta_1)$.

ii. Derive the maximum likelihood estimator $\widehat{\beta}_0, \widehat{\beta}_1$.

iii. Derive the distribution of $\widehat{\beta}_0$ and $\widehat{\beta}_1$. Based on this, suppose we want to test the hypothesis

$$H_0 : \widehat{\beta}_0 = 0 \text{ versus } H_1 : \widehat{\beta}_0 > 0,$$

write down the formula of the P-value. Similarly, suppose we want to test the hypothesis

$$H_0 : \widehat{\beta}_1 = 0 \text{ versus } H_1 : \widehat{\beta}_1 < 0,$$

write down the formula of the P-value.

(**Question3):** (45 points) Suppose we have a set of observations $(x_{i1}, x_{i2}, \cdots, x_{ip}, Y_i)$, assuming that it satisfies a linear model

$$Y_i = \sum_{j=1}^{p} x_{ij}\beta_j + \epsilon_i \text{ where } \epsilon_i \sim N(0, \sigma^2) \text{ and } \sigma \text{ is known.}$$

In a matrix form, we have

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon.$$

i. Write down the log-likelihood function, and prove the MLE for $\beta = (\beta_1, \cdots, \beta_p)^\top$ is equivalent to

$$(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}.$$

ii. Suppose we want to test the hypothesis

$$H_0 : \beta_1 = 0 \text{ versus } H_1 : \beta_1 > 0,$$

derive the formula of the P-value for this test. Hint: $\beta_1 = (1, 0, \cdots, 0)\beta$.

(**Question4):** (45 points) Suppose we have observations $(x_i, Y_i), i = 1, \cdots, n$. We want to fit a quadratic curve

$$Y_i = \beta_0 + \beta_1 \times x_i + \beta_2 \times x_i^2 + \epsilon_i, \text{ where } \epsilon_i \sim N(0, \sigma^2).$$

i. Write down the log-likelihood function and derive the MLE for $\widehat{\boldsymbol{\beta}}$ (matrix notation).

ii. Assume that σ^2 is known. Derive the $1 - \alpha$ confidence interval for β_1. (Hint: Your estimator should be a linear combination of Y_i, which has normal distribution).

iii. Suppose we want to test the hypothesis

$$H_0 : \beta_1 = 0, \text{ versus } H_1 : \beta_1 > 0,$$

establish the P-value of the test.

iii. For the data (6.91, 17.52) (4.32, 22.69) (2.38, 17.61) (7.98, 14.29) (8.26, 10.77) (2.00, 12.87) (3.10, 18.63) (7.69, 16.77) (2.21, 14.97) (3.42, 19.16) (8.18, 11.15) (5.39, 22.41) (1.19, 7.50) (3.21, 19.06) (5.47, 23.89) (7.35, 16.63) (2.32, 15.09) (7.54, 14.75) (1.27, 10.75) (7.33,17.42) (8.41, 9.40) (8.72, 9.83) (6.09, 22.33) (5.30, 21.37) (7.30, 17.36), assume $\sigma^2 = 1$, derive the 95% CI for β_1. Does it cover 0? What is the P-value of the test in iii.

2 Programming Questions

(40 points)

i. Generate i.i.d. data (X_i, Y_i) for $i = 1, 2, \cdots, n$, where $X_i \sim N(0, 1)$, $Y_i = X_i \times 0.5 + 0.1 + \epsilon_i$, and $\epsilon_i \sim N(0, 1)$. Estimate β_0 and β_1 using the formula in Question 2. Report the estimated β_0 and β_1.

ii. Repeat step i. for 200 times, and plot a histogram for $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

iii. Suppose we want to test the hypothesis

$$H_0 : \widehat{\beta}_1 = 0 \text{ versus } H_1 : \widehat{\beta}_1 < 0.$$

Implement the procedure for calculating P-value. Repeat this step for 200 times, plot the histogram of this P-value.